

Section B

7	(a)	The moment of a force about a pivot is the product of that force and the <u>perpendicular distance between the line of action of the force and the pivot.</u> The torque of a couple between two forces is the product of one of the forces and the <u>perpendicular distance between their lines of action.</u> <u>One force for moment & two forces for torque OR</u> <u>about a pivot for moment & no need for pivot for torque</u>	B1 B1 B1
	(b)(i)	The <u>anti-clockwise moment due to the weight of the hook is balanced by the total clockwise moments due to the weight of the sliding weights.</u> OR c.g of the weighing machine is at the pivot	B1
	(b)(ii)	<u>For the same load on the hook, a smaller perpendicular distance from the hook to the pivot will result in smaller anti-clockwise moment to be balanced</u> by the sliding weights. A longer rigid rod will allow for <u>smaller sliding weights to be used.</u> OR A longer rigid rod will give a better precision to the reading. OR With the <u>same sliding weights, a bigger range of loads can be measured.</u>	B1 B1
	(b)(iii)	Weight of sack of flour = mg $mg \times 4.8 = (12 \times 84) + (2.5 \times 72)$ $m = 25.2 \text{ kg}$	C1 A1
	(b)(iv)	The <u>movement of the sliding weights will be much smaller</u> for such a small load hence the <u>corresponding fractional uncertainty of the weight of the object will be higher.</u>	B1 B1
	(c)(i)	A force F acts on a mass m to move it vertically upwards <u>at constant speed</u> (so that no change in E_k) <u>by a displacement h in the direction of the force.</u> Since the object moves at constant speed, <u>the upward force, F, must be equal to the weight of the object, mg.</u> (no resultant force). Using <u>Work done on object = $Fh = (mg)h$</u> Since Fh is the work done on the object and is equal to the <u>increase</u> in potential energy, $E_p = mgh$	B1 B1 B1
	(c)(ii)1.	Power = $\frac{n \times mgh}{t} = \frac{60 \times (55)(9.81)(15)}{1 \times 30}$ $= 1.62 \times 10^4 \text{ W}$	C1 A1

	(c)(ii)2.	$v \text{ (in direction parallel to escalator)} = \frac{15}{\sin 30^\circ} \div 30 = 1.0 \text{ m s}^{-1}$ $\text{Power} = Fv = 1.0 \times 10^4 \times 1.0$ $= 1.0 \times 10^4 \text{ W}$	C1 C1 A1
	(c)(ii)3.	To maintain constant speed, the escalator has to supply <u>more power</u> to <u>overcome the (downward) force exerted on escalator steps by users</u> moving up. OR higher rate of increase of GPE of the users	B1 B1

8	(a)(i)	A spontaneous radioactive decay is one whereby it is not triggered by external	R1
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